

CUSTOMER CHURN PREDICTION SYSTEM

Professional Project Documentation (CEO & Industry Level)

1. Project Title

Customer Churn Prediction System using Machine Learning

2. Executive Summary

Customer retention is one of the most important business challenges across industries such as Telecommunications, Banking, Insurance, E-commerce, OTT Platforms, and SaaS companies.

This project uses Machine Learning algorithms to predict whether a customer is likely to leave the company (churn) or continue using its services. By identifying high-risk customers in advance, organizations can take proactive actions to improve retention and reduce revenue loss.

The system analyzes customer behavior, subscription details, service usage patterns, and billing information to generate churn predictions.

3. Business Problem Statement

Customer acquisition costs are significantly higher than customer retention costs. Losing customers directly impacts company revenue, market share, and long-term growth.

Current Challenges:

- High customer attrition rate
- Revenue loss due to churn
- Difficulty identifying dissatisfied customers
- Inefficient marketing campaigns
- Lack of proactive retention strategies

Proposed Solution:

Develop an AI-powered Customer Churn Prediction System that identifies customers who are likely to leave and enables the company to take preventive actions.

4. Project Objectives

The primary objectives of this project are:

- ✓ Predict customer churn accurately
- ✓ Identify factors influencing churn
- ✓ Reduce customer attrition
- ✓ Increase customer lifetime value
- ✓ Improve customer satisfaction
- ✓ Support data-driven decision making
- ✓ Enhance business profitability

5. Industry Applications

This solution can be implemented in:

Industry Use Case

Telecom Predict subscribers leaving network

Banking Predict account closures

Insurance Policy cancellation prediction

E-commerce Customer retention analysis

OTT Platforms Subscription cancellation

SaaS Companies User subscription renewal prediction

6. Real-World Scenario

Suppose a telecom company has:

- 10 Million customers
- Average monthly revenue = ₹500/customer

If 5% customers churn monthly:

Revenue Loss =

$$10,000,000 \times 5\% \times ₹500$$

$$= ₹250 \text{ Crores per month}$$

By implementing churn prediction and retaining even 20% of churn customers, the company can save crores of rupees annually.

7. Dataset Description

Dataset Size:

- Total Records: 30,000
- File Format: CSV

- Data Type: Structured Data

Features Description

Feature Description

CustomerID Unique customer identifier

Gender Male/Female

SeniorCitizen Senior citizen status

Partner Customer has partner or not

TenureMonths Duration of customer relationship

InternetService Internet service type

TechSupport Availability of technical support

PaperlessBilling Billing preference

PaymentMethod Payment mode

ContractType Customer contract type

MonthlyCharges Monthly bill amount

TotalCharges Total expenditure

Churn Target variable

8. Problem Type

Machine Learning Category:

Supervised Machine Learning

Learning Type:

Classification Problem

Target Variable:

Churn (Yes/No)

9. Machine Learning Workflow

Step 1: Data Collection

Collect customer data from:

- CRM systems
- Billing systems
- Customer service logs
- Usage databases

Step 2: Data Preprocessing

Operations Performed:

- ✓ Missing Value Handling
- ✓ Duplicate Removal
- ✓ Outlier Detection
- ✓ Data Cleaning

- ✓ Feature Transformation
- ✓ Data Standardization

Step 3: Exploratory Data Analysis (EDA)

Analyze:

- Customer distribution
- Churn percentage
- Monthly charges analysis
- Contract-wise churn
- Service-wise churn

Visualization Tools:

- Matplotlib
- Seaborn
- Plotly

10. Feature Engineering

Transform categorical variables into numerical values:

Encoding Techniques:

- Label Encoding
- One-Hot Encoding

Scaling Techniques:

- StandardScaler
- MinMaxScaler

11. Data Splitting

Dataset Split:

- Training Data = 80%
- Testing Data = 20%

Example:

24,000 records → Training

6,000 records → Testing

12. Machine Learning Algorithms Used

Algorithm Purpose

Logistic Regression Baseline model

Decision Tree Rule-based learning

Random Forest Ensemble learning

XGBoost Boosting technique

LightGBM High-speed boosting

SVM Hyperplane classification

KNN Similarity-based prediction

13. Model Training Process

1. Import Dataset

2. Preprocess Data

3. Encode Features

4. Split Dataset

5. Train Models

6. Evaluate Performance

7. Save Best Model

8. Deploy Model

14. Evaluation Metrics

Metric Formula

Accuracy $(TP+TN)/(TP+FP+TN+FN)$

Precision $TP/(TP+FP)$

Recall $TP/(TP+FN)$

F1 Score $2PR/(P+R)$

ROC-AUC Area under ROC curve

15. Confusion Matrix

Actual/Predicted Churn No Churn

Churn TP FN

No Churn FP TN

16. Expected Results

Expected Accuracy:

85% – 95%

Business Outcome:

- ✓ Reduced churn rate
- ✓ Increased customer retention
- ✓ Improved revenue generation
- ✓ Better customer satisfaction

17. Deployment Architecture

Customer Data



Data Preprocessing



Feature Engineering



Machine Learning Model



Prediction Engine



Risk Score Generation



Business Dashboard



Retention Action

18. Technology Stack

Component Technology

Programming Language Python

Data Processing Pandas, NumPy

Visualization Matplotlib, Seaborn

Machine Learning Scikit-Learn

Advanced ML XGBoost, LightGBM

Deployment Streamlit/Flask

Database MySQL

Version Control GitHub

19. Key Business KPIs

- Customer Retention Rate
- Churn Rate
- Customer Lifetime Value (CLV)
- Revenue Growth
- Customer Satisfaction Score
- Net Promoter Score (NPS)

20. Future Enhancements

- ✓ Real-time churn prediction
- ✓ Deep Learning integration
- ✓ Customer segmentation
- ✓ Recommendation engine
- ✓ Explainable AI (SHAP)
- ✓ Cloud deployment (AWS/Azure/GCP)
- ✓ Automated retention campaigns

21. Conclusion

The Customer Churn Prediction System leverages Machine Learning to identify customers at risk of leaving the organization. By enabling proactive retention strategies, businesses can significantly reduce revenue loss, improve customer satisfaction, and gain a competitive advantage. This project demonstrates how Artificial Intelligence can transform raw customer data into actionable business insights, supporting strategic decision-making at the executive level.

[Github Link](#)

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